Ungraded: Kernel PCA and MDS
Practice Assignment • 10 min

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Your grade: 100%

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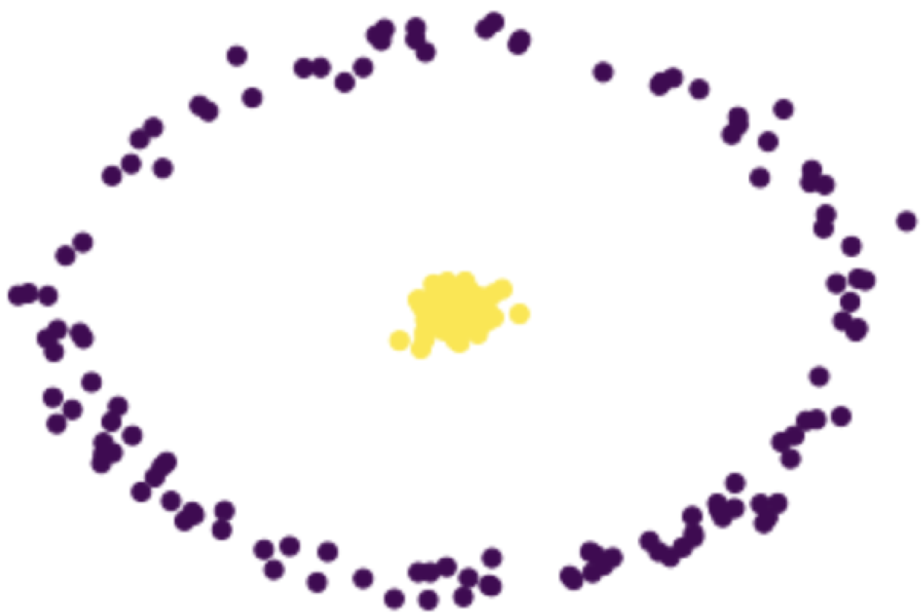
To pass you need at least 66%. We keep your highest score.

Next item →

1. Given the data visualized below with the classes represented by different colors, should PCA or kernel PCA be used, and why?

1 / 1 point



- ○ Neither because the data cannot be projected onto a lower dimension.
- ◉ Kernel PCA because the data is not linearly separable.

  Correct! If the data is projected directly onto a lower dimension, the different classes cannot be clearly separated with a single plane. Hence, we use a kernel function to map it to a higher dimension first, before applying PCA.

- ○ PCA because the data is clearly separable when projected onto a lower dimension.
- ○ Either is fine because the two classes are clearly separable.

2. How does the goal of MDS (Multidimensional Scaling) compare to PCA?

1 / 1 point

- ◉ MDS tries to maintain geometric distances between data points, whereas PCA tries to preserve variance within data.

  Correct! Although both are dimensionality reduction techniques, MDS will not strive to maintain the variance within the original data.

- ○ PCA tries to maintain geometric distances between data points, whereas MDS tries to preserve variance within data.
- ○ Both MDS and PCA try to preserve variance within data.
- ○ Both MDS and PCA try to to maintain geometric distances between data points.

3. (True/False) If the number of components is equal to the dimension of the original features, kernel PCA will reconstruct the data, returning the original.

1 / 1 point

- ◉ False

  Correct! Kernel PCA spans a subspace of the original data, so applying inverse transformation on the data after kernel PCA will not return the original data. You can review Practice lab: Kernel PCA for more information.

- ○ True